

Coherent Optical Frequency Transfer via a 490 km Noisy Fiber Link

Xiang Zhang(张翔)^{1,2,3}, Xue Deng(邓雪)^{1,3}, Qi Zang(臧琦)^{1,2,3}, Dongdong Jiao(焦东东)^{1,3}, Jing Gao(高静)^{1,2,3}, Dan Wang(王丹)^{1,2,3}, Qian Zhou(周茜)^{1,2,3}, Jie Liu(刘杰)^{1,3}, Guanjun Xu(许冠军)^{1,3}, Ruifang Dong(董瑞芳)^{1,2,3*}, Tao Liu(刘涛)^{1,2,3*}, and Shougang Zhang(张首刚)^{1,2,3}

¹National Time Service Center, Chinese Academy of Sciences, Xi'an 710600, China

²University of Chinese Academy of Sciences, Beijing 100039, China

³Key Laboratory of Time and Frequency Standards, Chinese Academy of Sciences, Xi'an 710600, China

(Received 18 December 2021; accepted 16 February 2022; published online 15 March 2022)

We demonstrate the coherent transfer of an ultrastable optical frequency reference over a 490 km noisy field fiber link. The fiber-induced phase noise power spectrum density per-unit-length at 1 Hz offset frequency can reach up to $510 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$, which is much higher than the fiber noise observed in previous reports. This extreme level of phase noise is mainly due to the fiber link laying underground along the highway. Appropriate phase-locked loop parameters are chosen to complete the active compensation of fiber noise by measuring the intensity fluctuation of additional phase noise and designing a homemade digital frequency division phase discriminator with a large phase detection range of $2^{12}\pi$ rad. Finally, a noise suppression intensity of approximately 40 dB at 1 Hz is obtained, with fractional frequency instability of 1.1×10^{-14} at 1 s averaging time, and 3.7×10^{-19} at 10000 s. The transfer system will be used for remote atomic clock comparisons and optical frequency distribution over a long-distance communication network established in China.

DOI: 10.1088/0256-307X/39/4/044201

Recently, with the realization of 10^{-18} level accuracy and instability of the optical clock,^[1–4] high-precision frequency standards have played a significant role in various fields of research, such as radio astronomy,^[5,6] fundamental physics and applied research,^[7,8] chronometric geodesy,^[9] gravitational wave detection^[10] and searching for dark matter.^[11]

To complete the frequency comparison of remote optical clocks and to achieve long-distance frequency reference distribution, high-precision optical frequency transfer via communication fiber has received extensive attention and has been thoroughly investigated.^[12–21] One critical factor limiting system performance of long-distance frequency dissemination is the free-running phase noise induced by environmental perturbations along with the fiber link, which is mainly determined by the routing solution of the communication fiber links. Active noise cancellation (ANC) scheme has been proposed and demonstrated by Ma *et al.*^[22] compensating for additional phase fluctuation. The fiber-induced phase noise is detected in the configuration by beating the local optical reference and the round-trip optical signal, and it is compensated successfully using a conventional phase-locked loop (PLL). To date, massive coherent transfer experiments of an optical frequency over underground and submarine fiber links with lower-level phase-noise power spectrum density (PSD)^[23] have

been successfully demonstrated. Especially, the European metrological network of 4802 km coherent optical fiber links has been established and reported, and 2198 km fiber-based multiuser optical frequency dissemination with uncertainties below 1.1×10^{-19} have been demonstrated and steadily operated for two years.^[24] However, noisy fiber links with high-level-phase-noise PSD, such as aerial suspended fibers^[25] or underground fiber links laying along speedway,^[26] can cause a false phase identification and increase phase-slip rate in the transfer system, deteriorating the long-term transfer instability of ultrastable optical frequency, as described in Ref. [25]. Meanwhile, considering that most of the communication fiber links established in China have a higher-intrinsic-phase-noise PSD because they are assembled underground along the speedway, it is necessary to design a coherent optical frequency transfer system with a large phase detection range for the noisy fiber links.^[27]

In this study, we present optical frequency transfer via a 490 km noisy fiber implemented in China. Seven bidirectional erbium-doped fiber amplifiers (Bi-EDFAs) are along with this fiber link to compensate for the power attenuation of transmitted light. The phase-noise PSD per-unit-length at 1 Hz offset frequency of the employed fiber link reaches up to $510 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$, which is much higher than the fiber noise observed in previous reports, such

*Corresponding authors. Email: taoliu@ntsc.ac.cn; dongruifang@ntsc.ac.cn
© 2022 Chinese Physical Society and IOP Publishing Ltd

as $4 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$ (USA),^[12] $0.5 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$ (Germany),^[28] $3 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$ (France),^[16,29] $20 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$ (Italy),^[15] $20 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$ (Japan).^[30] Though a higher phase-noise PSD was reported previously,^[31] the integrated phase noise after transmitting through the 490 km field fiber is significantly large. Compared with the 1840 km fiber link,^[14] the integrated phase noise in our system is significantly higher. A programmable digital frequency division phase discriminator (PDFD) based on the ADF4002 frequency synthesizer is designed to efficiently suppress the large phase noise, which

has a large phase detection range up to $2^{12}\pi$ rad. Frequency division parameters of 1280 are used to accurately identify the phase fluctuation of the transmitted light and complete the noise compensation of fiber noise, which ensures that the integrated phase noise of the obtained RF signal in the PLL is less than 1 rad. Noise suppression of 40 dB at 1 Hz is obtained via active phase noise cancelation, and a transfer frequency instability of 1.1×10^{-14} at 1 s and 3.7×10^{-19} at 10000 s are achieved, which are sufficient for comparing the most advanced optical clocks in China to date.

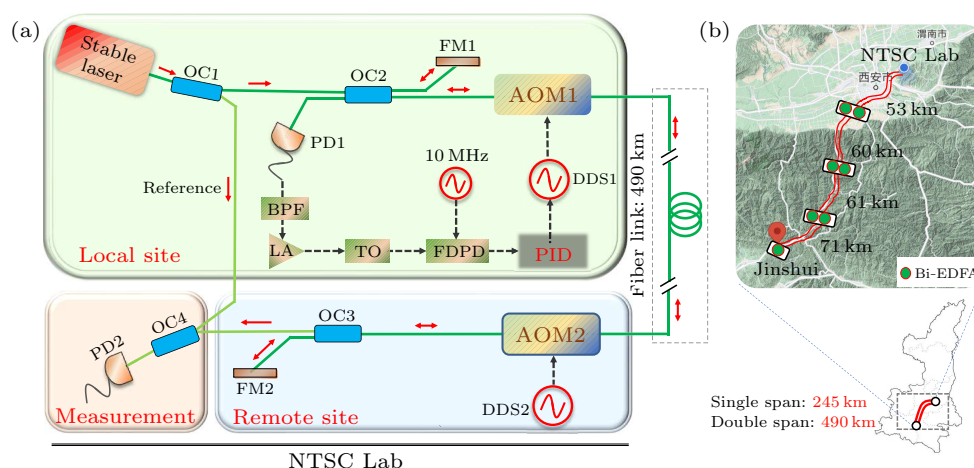


Fig. 1. (a) Experimental setup for optical frequency transfer over a 490 km noisy fiber link using active noise control (ANC) technology, optical coupler (OC), an acoustic-optic modulator (AOM), Faraday mirror (FM), frequency division phase discriminator (FDPD), photodetector (PD), a band-pass filter, and logarithmic amplifier (LA). The green solid arrows represent the direction of the RF signals. (b) The map of the flied fiber link provided by communication operators is buried along the highway. The length of the single-span fiber from NTSC to Jinshui is approximately 245 km.

Figure 1(a) depicts the experimental setup of coherent optical frequency transfer with the ANC technology. The frequency transfer system, similar to the experimental apparatus demonstrated in Ref. [29] has two optical sites, namely, local and remote sites. At the local site, the line width of the commercial CW fiber laser (NKT E15) is reduced to below 1 Hz by locking it to an ultrastable optical reference cavity.^[32] The transmitted light has a central frequency of 193.4 THz, and its power is further split into two portions with a 90/10 OC. The large proportion of transmitted light passes through an AOM1 driven by a 110 MHz radio-frequency (RF) signal deriving from a direct digital synthesis, which is then transmitted to a remote site through the 490 km fiber link. At the remote site, the transmitted light is connected to AOM2 driven by a 50 MHz RF signal, which is applied to discriminate the wanted round-trip light signal from the stray reflections between fiber connectors and splices. Seven Bi-EDFAs are installed along with the fiber link to compensate for the power attenuation of the transmitted light. The gain of each amplifier is limited to less than 16 dB

to prevent the self-oscillation phenomenon between Bi-EDFAs. This active compensation system is, in essence, a Michelson interferometer, with the 490 km fiber link serving as the interferometer's long arm and the light reflected by Faraday mirror (FM1) serving as the reference light. The photodetector 1 (PD1) detects the heterodyne beat signal 320 MHz between the round-trip light reflected by FM2 and the reference light. After passing through an RF band-pass filter (4 MHz bandwidth) and low-noise amplifier (LNA 605) [Fig. 1(a)], the obtained RF signal is sent to a tracking oscillator filter (~ 100 kHz bandwidth). The output 320 MHz signal is then sent to a programmable digital PDFD with a 10 MHz reference signal that is derived from a hydrogen maser to derive the phase error. A schematic of the programmable PDFD is shown in Fig. 2(a). The frequency division parameters N and M are set by the microcontroller, resulting in a phase comparison between two RF signals with the same frequency. Here, with the noisy 490 km fiber link, the parameters N and M are chosen as 40 and 1280, respectively, to effectively identify the phase fluctuation. The phase error signal is applied to servo DDS1

through a PI controller after passing through a low filter and Adder. The phase noise attached to transmitted light will be actively eliminated using the ANC scheme within the control bandwidth. The phase error shows a linear output when 320 MHz RF signal is derived from RF signal generator (SG382) based on jagged phase modulation, as shown in Fig. 2(b). The actual phase error when the 490 km fiber link is under phase-uncompensated condition is shown in Fig. 2(c). Finally, the resulting out-of-loop RF signal of 160 MHz detected by PD2 is recorded using a deadtime free frequency counter (K&K FXE) operating at Λ -mode to calculate the frequency instability.

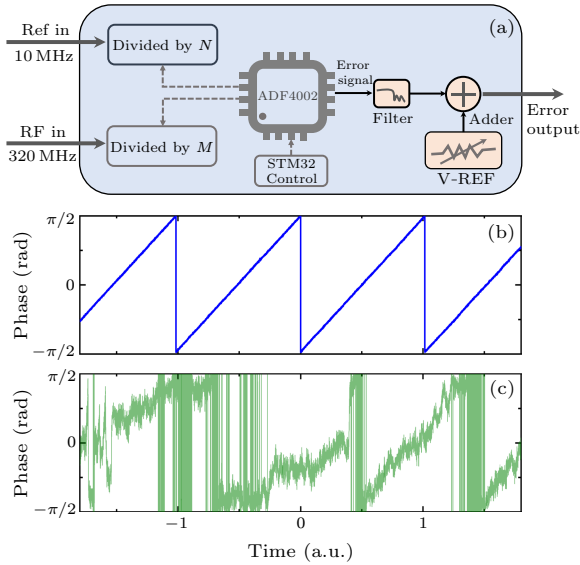


Fig. 2. (a) Simplified block diagram of the programmable digital FDPD based on the ADF4002 frequency synthesizer. (b) The phase error output from the FDPD when the 320 MHz RF signal is derived from the RF signal generator (SG382) based on jagged phase modulation. (c) The phase error output from the FDPD when the 490 km fiber link is under phase-uncompensated condition, the parameters N and M are set as 40 and 1280, respectively.

In the process of active noise suppression of the 490 km noisy fiber link, the overall performance of the phase noise rejection capability significantly depends on the proper setting of the loop parameters. The most important thing is to determine the appropriate frequency division parameter. Therefore, we measured the phase-noise PSDs of the transferred light without noise compensation (black curve) [Fig. 3(b)]. The free-running phase-noise PSD of this employed 490 km fiber link approximately follows a power-law relationship, $S_{\text{free}}(f) \simeq h_0 f^{-2} \text{ rad}^2/\text{Hz}$, for the Fourier frequency below approximately 10 Hz, indicating that the white frequency noise is dominant under the noise uncompensated condition. The h_0 is up to approximately 250000 in this study, indicating that the phase noise per unit length of this noisy fiber link at 1 Hz is $510 \text{ rad}^2 \cdot \text{Hz}^{-1} \cdot \text{km}^{-1}$. Such a large amount of fiber noise is mainly attributed to the heavy traffic along the highway. Additionally, the

uncompensated phase-noise PSD shows a broad peak at the frequency between 10 and 20 Hz, which has been reported as similar behavior in previous results and has been attributed to the building and ground vibrations.^[15] The integral phase jitter is approximately 720 rad for the free-running link [Fig. 3(a)]. A frequency division factor of 1280 is finally selected and used to complete the phase noise compensation to correctly identify the phase fluctuation of the transmitted light.

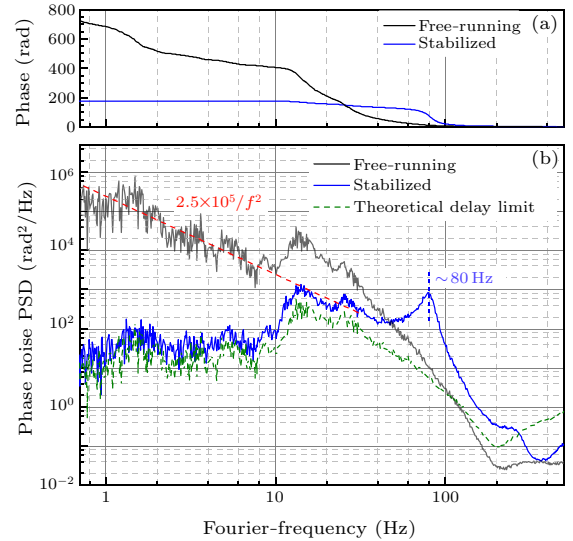


Fig. 3. (a) The calculated phase jitter in the free-running (black curve) and stabilized (blue curve) fiber link. (b) The measured phase-noise PSDs in the free-running case (black curve) and stabilized case (blue curve). The servo bump is approximately 80 Hz because of the propagation delay is shown. The green dotted curve represents the theoretical compensation limitation according to Ref. [12].

After actively compensating the fiber-induced phase fluctuation, the phase-noise PSD reduction at 1 Hz achieves approximately 40 dB, and the residual-phase-noise PSD $S_r(f)$ illustrates a dependence with a lower Fourier frequency of $h_1 f^0$, showing that the remaining noise is mainly determined by the white phase noise in the stabilized 490 km fiber link. Here, $h_1 \simeq 20$, and the integral phase jitter is approximately 170 rad for the stabilized link (Fig. 3). Because of the limitation induced by the delay $\tau = nL/c$ of the fiber link, with $n = 1.468$ representing the refractive index of the fiber link, c representing the speed of light in vacuum, and $L = 490 \text{ km}$, a servo bump at 80 Hz in the noise-compensated case is observed, which is close to the calculated servo bandwidth ($\sim 1/4\tau$) of 106 Hz. The residual phase-noise PSD $S_r(f)$ agrees well with the theoretical compensation limitation of the phase-noise PSD [Fig. 3(b)], as shown in Ref. [11],

$$S_{\text{com}}(f) = \frac{1}{3} (2\pi f \tau)^2 S_{\text{free}}(f). \quad (1)$$

During the link operation, we measure the end-to-end phase data using a Λ -type frequency counter with

1 s gate time, the amplitude fluctuations of 160 MHz beat note signal is less than 1 dB. The effective suppression of the fiber-induced phase fluctuation in the time domain is shown in Fig. 4, and no cycle slips were observed during the phase acquisition. The long-term phase fluctuation could reach 300000 optical cycles peak to peak of over approximately 33000 s for the free-running 490 km fiber link, which is mainly attributed to the temperature fluctuation and traffic disturbance along with the noise fiber link. After being compensated, the end-to-end phase drift is well suppressed to approximately 1.2 peak-to-peak optical cycles.

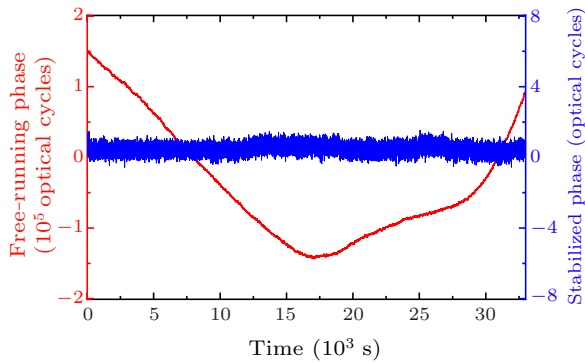


Fig. 4. The temporal phase fluctuation of the 490 km fiber link in the free-running case (red curve, left axis) and stabilized case (blue curve, right axis). Data were measured with a deadline free Λ -type frequency counter with 1 s gate time.

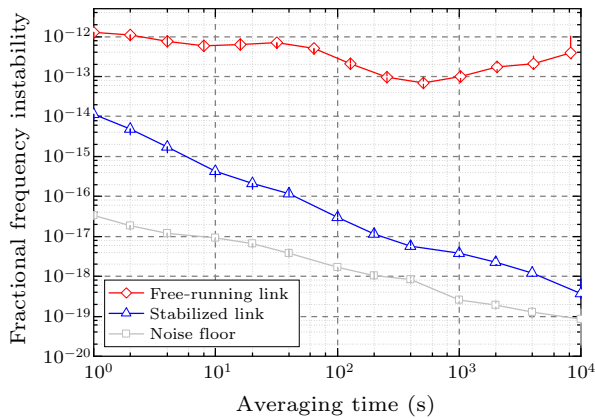


Fig. 5. The measured fractional frequency instability in terms of Mod-ADEV for the free-running (red curve) and stabilized (blue curve) fiber link. Furthermore, the measured noise floor of the interferometer (gray curve) is shown. The error bars are inside the symbols.

Figure 5 illustrates the fractional frequency instability in terms of the Modified Allan deviation (Mod-ADEV), which is calculated from the recorded frequency data. For the free-running, 490 km fiber link, the instability is 1.2×10^{-12} at the averaging time of 1 s and fluctuates around the 10^{-13} level (red curve). With the implementation of the fiber noise cancellation, optical frequency transfer achieves a fractional frequency instability of 1.1×10^{-14} at an integration

time of 1 s, and falls off with a slope of $\tau^{-3/2}$ for averaging times of up to 100 s, showing the characteristic response of the white phase noise. Finally, a transfer instability of 3.7×10^{-19} at 10000 s is achieved. As described in Ref. [12], the fractional frequency instability at 1 s averaging time σ_{1s} in the stabilized fiber link could be predicted as

$$\sigma_{1s} = \sqrt{\frac{8h_0}{3}} \frac{\tau}{\nu_0}, \quad (2)$$

where ν_0 is the optical carrier frequency. For the noisy 490 km fiber link, the delay $\tau \simeq 2.4$ ms and the σ_{1s} is calculated to be 9.1×10^{-15} based on Eq. (2), which agrees well with the experimental value of 1.1×10^{-14} (Fig. 5). The noise floor of the fiber interferometer is also measured using a 30 cm short-cutting stabilized fiber link, which is 3.4×10^{-17} at 1 s and reaches a floor in the 10^{-20} range at 10000 s averaging time. We attribute the variation of the long-term instability to the temperature fluctuations of the asymmetric optical fiber in the local and remote sites, where the active temperature controller is not used.

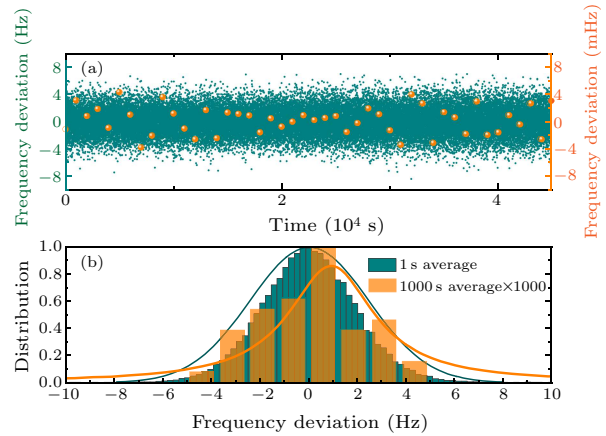


Fig. 6. (a) Frequency comparison between input and retrieved laser after a phase-compensated 490 km fiber link. A deadline free Λ -type frequency counter with a gate time of 1 s (dark cyan points, left axis) was used to record 45000 data points. The unweighted arithmetic means for all cycle-slip free 1000 s long segments were calculated, resulting in 45 data points (orange points, right frequency axis, enlarged scale). (b) Histograms and Gaussian fit for the recorded frequency values and above-mentioned 45 arithmetic mean values, respectively.

The accuracy is further evaluated by analyzing the end-to-end beat note frequencies. The frequency deviation of the retrieved light after actively stabilizing 490 km fiber link is shown in Fig. 6(a), which is continuously recorded using a Λ -type frequency counter with a 1 s gate time, resulting in successive 45000 points (dark cyan points, left axis). By binning them into $N = 45$ groups with each group having 1000 points, the arithmetic means of all cycle-slip free 1000 s intervals (orange points, right axis) is calculated as shown in Fig. 6(a). The Histograms and Gaussian fits for the frequency deviations are illustrated in Fig. 6(b). The

results show that the mean frequency is shifted by 0.19 mHz, which corresponds to a fractional value of 9.7×10^{-19} . The standard deviation is 2 mHz, with a corresponding fractional value of $\sigma_{1000} \sim 1.0 \times 10^{-17}$, which is a factor of 1000 less than the frequency instability at 1 s as expected for this Λ -type evaluation. Finally, the accuracy of the retrieved optical signal at the remote site is conservatively achieved 1.4×10^{-18} ($\sigma = \sigma_{1000}/\sqrt{N}$).

In summary, we have demonstrated a coherent optical frequency transfer system over a 490 km noisy fiber link buried along the highway. Faced with the high level of fiber-induced phase-noise PSD, a flexible and robust PDFD based on the ADF4002 frequency synthesizer is designed, which can provide a large phase detection range up to $2^{12}\pi$ rad. The active phase compensation achieves a noise suppression of 40 dB at 1 Hz, and fractional frequency instability of 1.1×10^{-14} at 1 s and 3.7×10^{-19} at 10000 s. The system would be suitable for long-distance and noisy-fiber-based optical frequency transmission and would be used for remote optical clocks comparisons in China.

Acknowledgments. This work was supported by the National Key Research and Development Program of China (Grant No. 2016YFF0200200), the National Natural Science Foundation of China (Grant Nos. 91636101, 12103059, 91836301, and 11803041), the Youth Innovation Promotion Association of the Chinese Academy of Sciences (Grant No. 1188000XGJ), the West Light Foundation of the Chinese Academy of Sciences (Grant No. XAB2016B47), and the Strategic Priority Research Program of the Chinese Academy of Sciences (Grant No. XDB21000000).

References

- [1] Ludlow A D, Boyd M M, Ye J, Peik E and Schmidt P O 2015 *Rev. Mod. Phys.* **87** 637
- [2] Marti G E, Hutson R B, Goban A, Campbell S L, Poli N and Ye J 2018 *Phys. Rev. Lett.* **120** 103201
- [3] Schioppo M, Brown R C, McGrew W F, Hinkley N, Fasano R J, Beloy K, Yoon T H, Milani G, Nicolodi D, Sherman J A *et al.* 2017 *Nat. Photon.* **11** 48
- [4] McGrew W F, Zhang X, Fasano R J, Schäffer S A, Beloy K, Nicolodi D, Brown R C, Hinkley N, Milani G, Schioppo M, Yoon T H and Ludlow A D 2018 *Nature* **564** 87
- [5] Clivati C, Ambrosini R, Artz T, Bertarini A, Bortolotti C, Frittelli M, Levi F, Mura A, Maccaferri G, Nanni M *et al.* 2017 *Sci. Rep.* **7** 40992
- [6] Wang B, Zhu X, Gao C, Bai Y, Dong J W and Wang L J 2015 *Sci. Rep.* **5** 13851
- [7] Lisdat C, Grosche G, Quintin N, Shi C, Raupach S M, Grebing C, Nicolodi D, Stefani F, Al-Masoudi A, Dörscher S *et al.* 2016 *Nat. Commun.* **7** 12443
- [8] Hu L, Poli N, Salvi L and Tino G M 2017 *Phys. Rev. Lett.* **119** 263601
- [9] Grotti J, Koller S, Vogt S, Häfner S, Sterr U, Lisdat C, Denker H, Voigt C *et al.* 2018 *Nat. Phys.* **14** 437
- [10] Kolkowitz S, Pikovski I, Langellier N, Lukin M D, Walsworth R L and Ye J 2016 *Phys. Rev. D* **94** 124043
- [11] Roberts B M, Delva P, Al-Masoudi A, Amy-Klein A, Bærentsen C, Baynham C F A, Benkler E, Bilicki S, Bize S, Bowden W, Calvert J, Cambier V, Cantin E, Curtis E A, Dörscher S, Favier M, Frank F, Gill P, Godun R M, Grosche G, Guo C, Hees A, Hill I R, Hobson R, Huntemann N, Kronjäger J, Koke S, Kuhl A, Lange R, Legero T, Lipphardt B, Lisdat C, Lodewyck J, Lopez O, Margolis H S, Alvarez-Martínez H, Meynadier F, Ozimek F, Peik E, Pottie P E, Quintin N, Sanner C, Sarlo L D, Schioppo M, Schwarz R, Silva A, Sterr U, Tamm C, Targat R L, Tuckey P, Vallet G, Waterholter T, Xu D and Wolf P 2020 *New J. Phys.* **22** 093010
- [12] Williams P A, Swann W C and Newbury N R 2008 *J. Opt. Soc. Am. B* **25** 1284
- [13] Lopez O, Haboucha A, Chanteau B, Chardonnet C, Amy-Klein A and Santarelli G 2012 *Opt. Express* **20** 23518
- [14] Droste S, Ozimek F, Udem T W, Predehl K, Hänsch T, Schnatz H, Grosche G and Holzwarth R 2013 *Phys. Rev. Lett.* **111** 110801
- [15] Calonico D, Bertacco E K, Calosso C E, Clivati C, Costanzo G A, Frittelli M, Godone A, Mura A, Poli N, Sutyrin D V *et al.* 2014 *Appl. Phys. B* **117** 979
- [16] Chiodo N, Quintin N, Stefani F, Wiotte F, Camisard E, Chardonnet C, Santarelli G, Amy-Klein A, Pottie P E and Lopez O 2015 *Opt. Express* **23** 33927
- [17] Deng X, Liu J, Jiao D D, Gao J, Zang Q, Xu G J, Dong R F, Liu T and Zhang S G 2016 *Chin. Phys. Lett.* **33** 114202
- [18] Wu L, Jiang Y, Ma C, Yu H, Bi Z and Ma L 2016 *Opt. Lett.* **41** 4368
- [19] Feng Z, Zhang X, Wu R, Sun Y, Wei F, Yang F, Gui Y and Cai H 2019 *Photon. Res.* **13** 1
- [20] Hu L, Tian X, Wu G and Chen J 2020 *Opt. Lett.* **45** 4308
- [21] Husmann D, Bernier L G, Bertrand M, Calonico D, Chaloulos K, Clausen G, Clivati C, Faist J, Heiri E, Hollenstein U, Johnson A, Mauchle F, Meir Z, Merkt F, Mura A, Scalari G, Scheidegger S, Schmutz H, Sinhal M, Willitsch S and Morel J 2021 *Opt. Express* **29** 24592
- [22] Ma L S, Jungner P, Ye J and Hall J L 1994 *Opt. Lett.* **19** 1777
- [23] Clivati C, Tampellini A, Mura A, Levi F, Marra G, Galea P, Xuereb A and Calonico D 2018 *Optica* **5** 893
- [24] Cantin E, Tønnes M, Targat R L, Amy-Klein A, Lopez O and Pottie P E 2021 *New J. Phys.* **23** 053027
- [25] Gozzard D R, Schediwy S W, Wallace B, Gamatham R and Grainger K 2017 *Opt. Lett.* **42** 2197
- [26] Zhang X, Hu L, Deng X, Zang Q, Liu J, Wang D, Liu T, Dong R and Zhang S 2021 arXiv:2106.12897 [physics.ins-det]
- [27] Walls C F M L F, Clements A and Vanek M 1990 *National Institute of Standards and Technology (NIST) Technical Note* 1337
- [28] Raupach S M, Koczwara A and Grosche G 2014 *Opt. Express* **22** 26537
- [29] Lopez O, Haboucha A, Kéfélian F, Jiang H, Chanteau B, Roncin V, Chardonnet C, Amy-Klein A and Santarelli G 2010 *Opt. Express* **18** 16849
- [30] Akatsuka T, Goh T, Imai H, Oguri K, Ishizawa A, Ushijima I, Ohmae N, Takamoto M, Katori H, Hashimoto T *et al.* 2020 *Opt. Express* **28** 9186
- [31] Fujieda M, Kumagai M and Nagano S 2010 *IEEE Trans. Ultrason. Ferroelectr. Freq. Control.* **57** 168
- [32] Jiao D, Gao J, Deng X, Xu G, Liu J, Liu T, Dong R and Zhang S 2020 *Opt. Commun.* **463** 125460